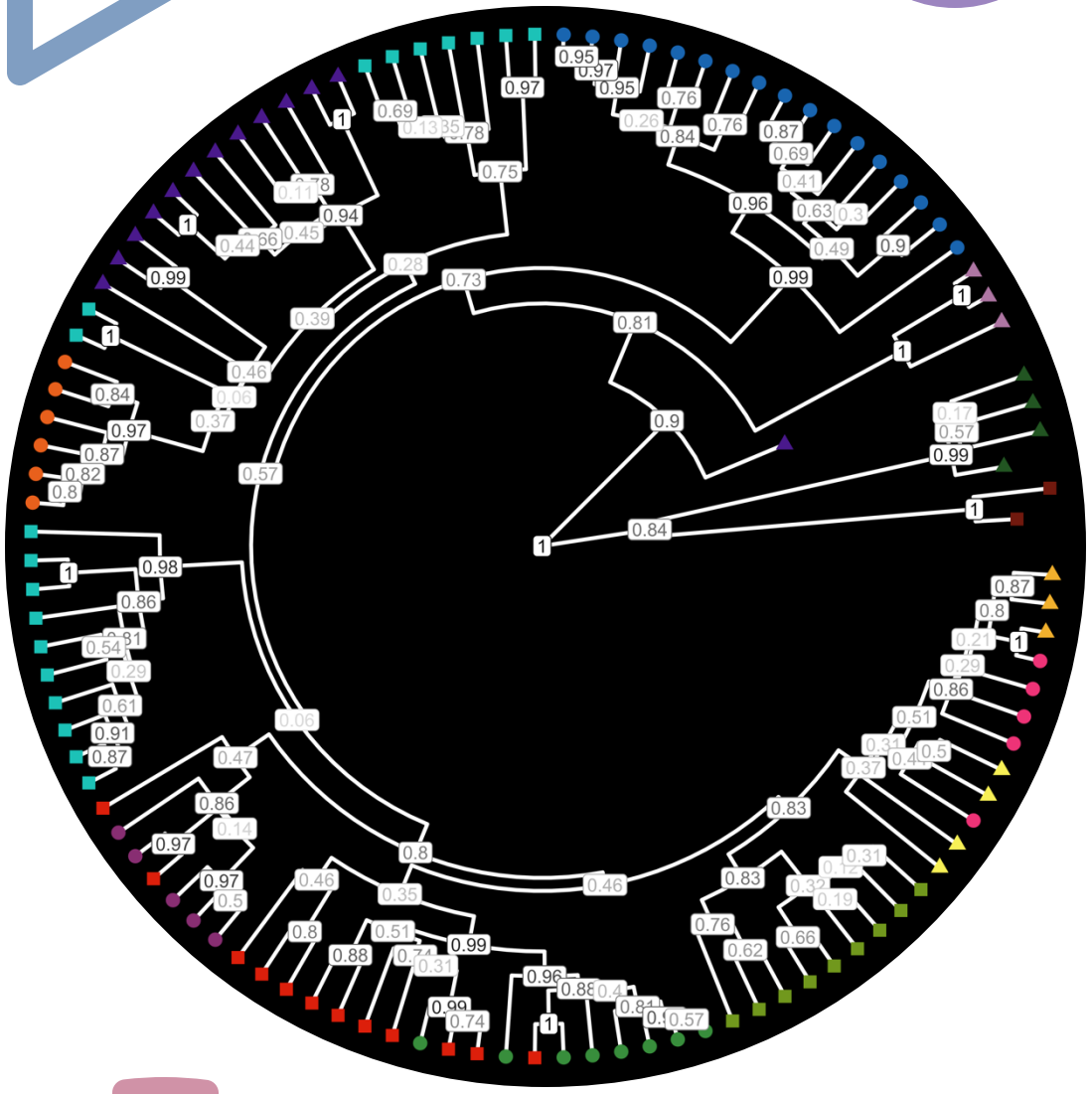
UNIVERSITÄT BERN

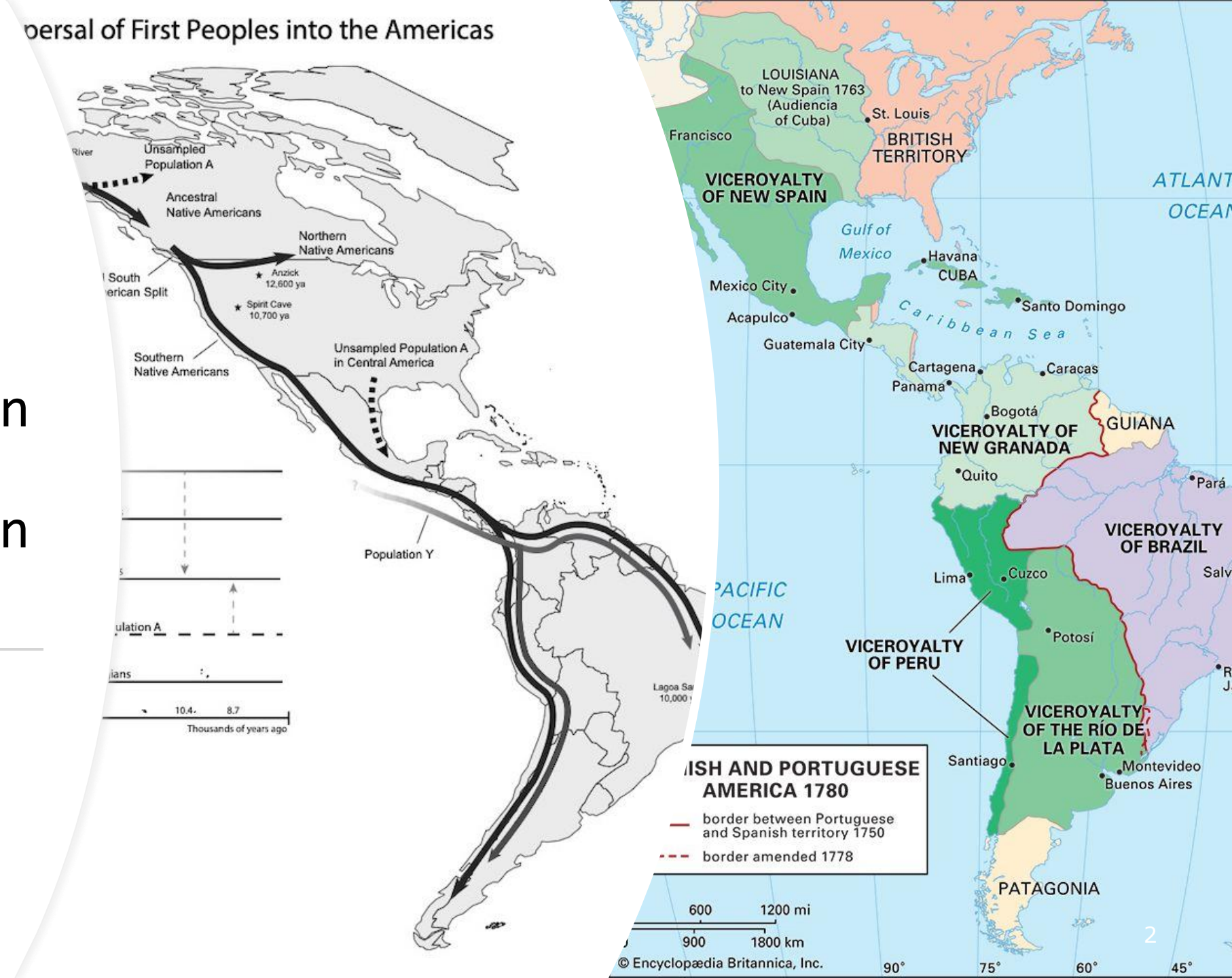
Exploring diversification through integrated language comparison



Sandra Auderset
University of Bern
sandra.auderset@unibe.ch

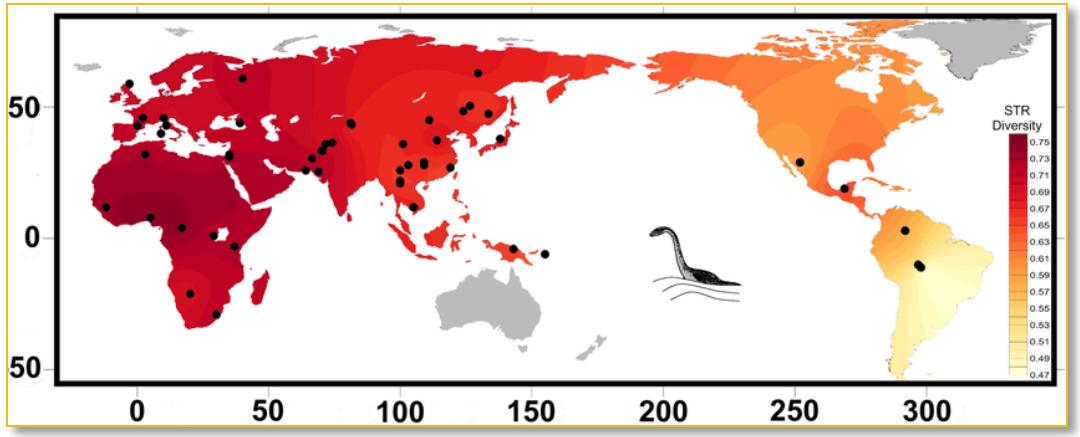
ICHL 27
Santiago de Chile
18.08.2025

Diversification and Colonization in the Americas



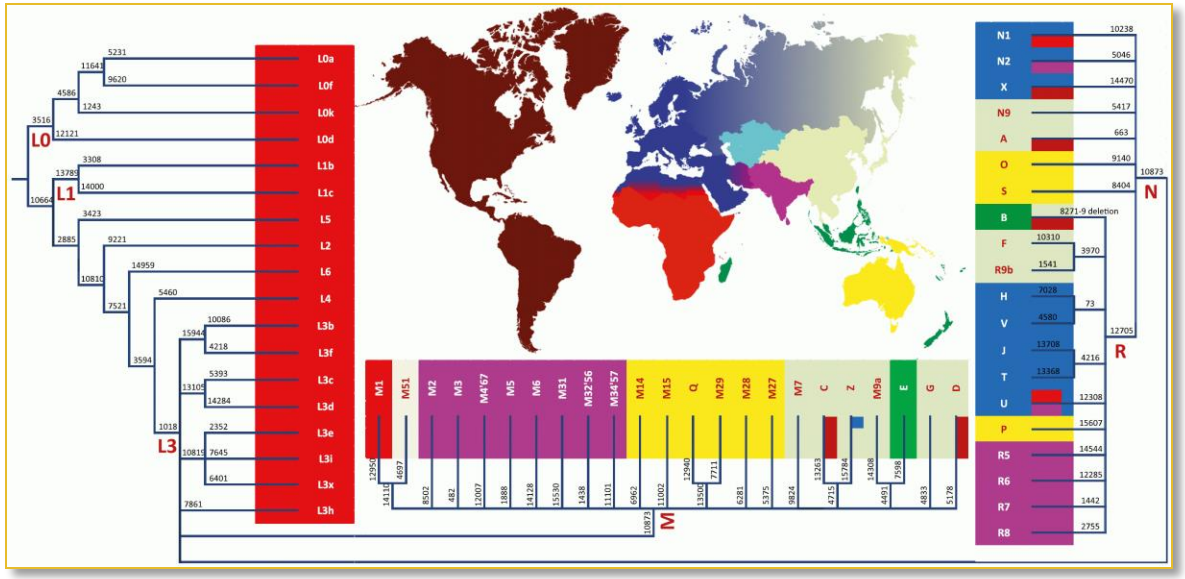
Diversification in the Americas

Genetics



pattern of genetic diversity in worldwide human populations (Pagani 2013)

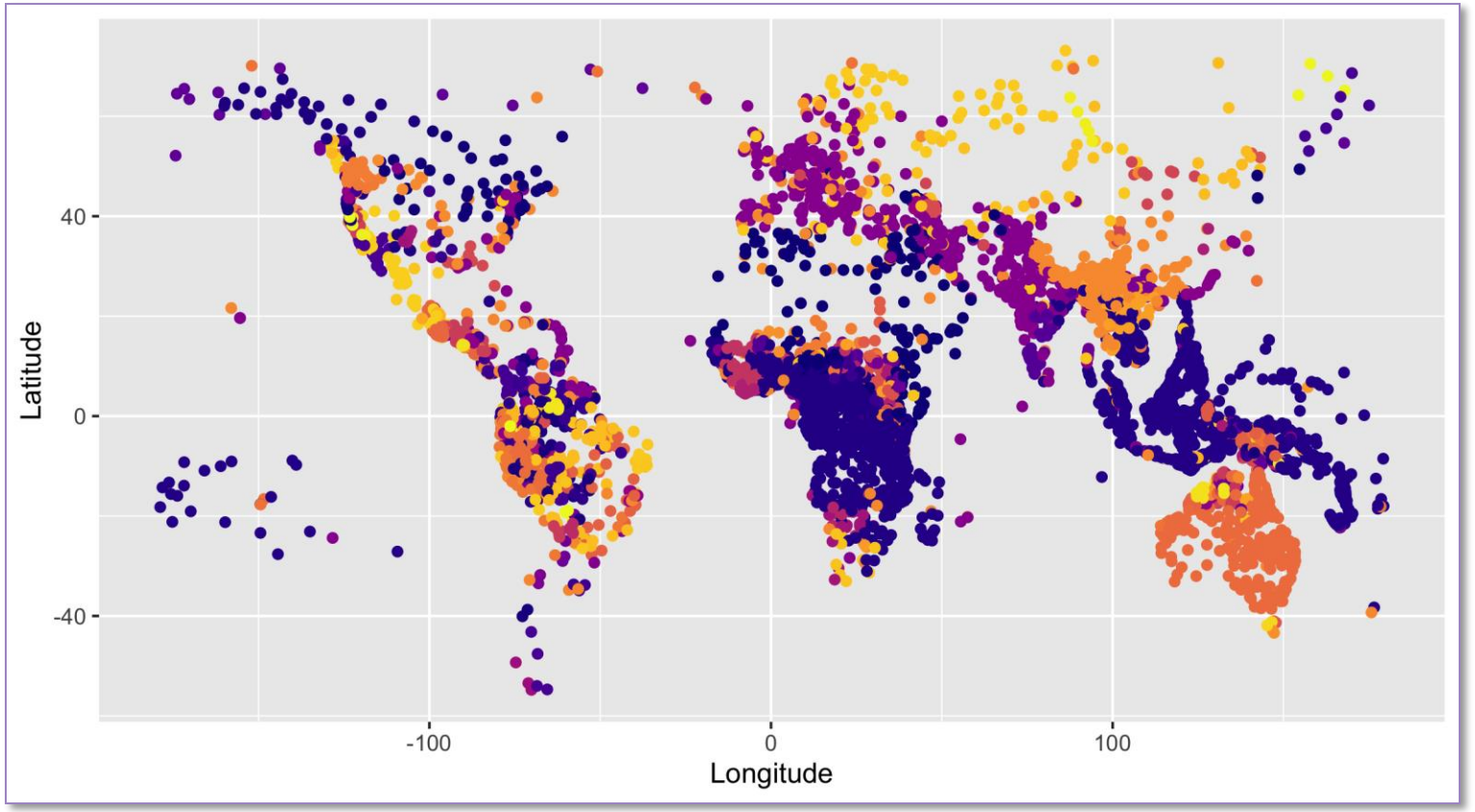
mtDNA haplogroup tree and map (Kivisild 2015)



Pagani, L. 2013. *Through the layers of the Ethiopian genome: a survey of human genetic variation based on genome-wide genotyping and re-sequencing data*. U Cambridge PhD Thesis | Kivisild, T. 2015. Maternal ancestry and population history from whole mitochondrial genomes. *Investigative Genetics* 6:3

Diversification in the Americas

Linguistics



Languages of the world colored by family membership (based on Glottolog)

Diversification in the Americas

Nichols (1990):

- Americas are linguistically very diverse
  - density of lineages
  - elaboration of lineages
  - typological diversity
- this linguistic diversity is too great for late population around 12’000 years ago
- more reasonable assumption would be 20’000-40’000 years ago

Nettle (1999):

- assumption of a linear increase of stocks over time is not warranted
- his model predicts an increase of stocks when an area is first populated, but the more time passes the more the rate decreases
- so the linguistic diversity in the Americas could be reconciled with a late population

if we understand linguistic diversification better, we might be able to say more about population and migration history of the Americas

Nichols, J. 1990. Linguistic diversity and the first settlement of the New World. *Language, 66(3)* | Nettle, D. 1999. Is the rate of linguistic change constant?. *Lingua, 108(2-3)*

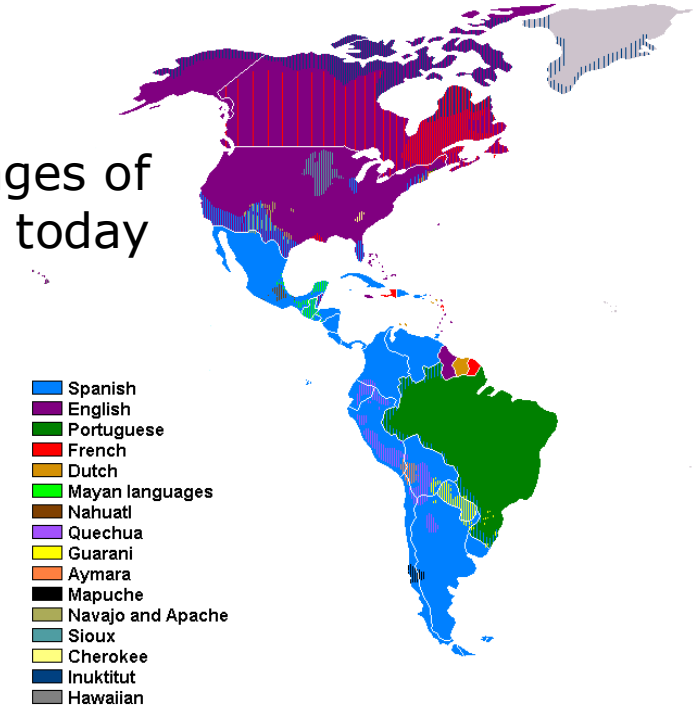
Diversification and Colonization

"The colonial period resulted in the enormous growth of a few languages. (…) In many areas, indigenous languages have been preserved in smaller pockets with a relatively high amount of speakers." (Carling 2024: 45)



Mesoamerican languages "pre-contact"

major languages of the Americas today



Carling, G. 2024. *Linguistic archaeology: An introduction and methodological guide*

https://commons.wikimedia.org/wiki/File:Pre-contact_distribution_language_families_Mexico.svg

https://gifex.com/fullsize-en/2009-11-13-11079/Languages_of_the_Americas.html

Diversification and Colonization in Quechua

traditional model with Pre-Inca diversification

Quechua I:	Huanca, Ancash, Huallaga	
Quechua II:	Quechua IIA:	Cajamarca, Incahuasi, Yauyos
	Quechua IIB:	Chachapoyas-Lamas, Quechua of D. de Santo Tomás, **Ecuadorian Quechua**
	Quechua IIC:	Ayacucho, Cuzco-Bolivian [incl. Imperial Incan], Argentina

"The linguistic and historical evidence points to an account of EQ's origin as Inca Quechua adopted and spread by the Spanish, and not as pre-Inca Quechua later subject to "Cuzco superstrate". EQ's rapid diversification into four languages since the colonial period might imply a similar chronology for the diversification of other Quechuas (…)." (Floyd 2024: 38-39)

Floyd, S. 2024. Ecuadorian Quechua and Quechuan classification. *Journal of Historical Linguistics*

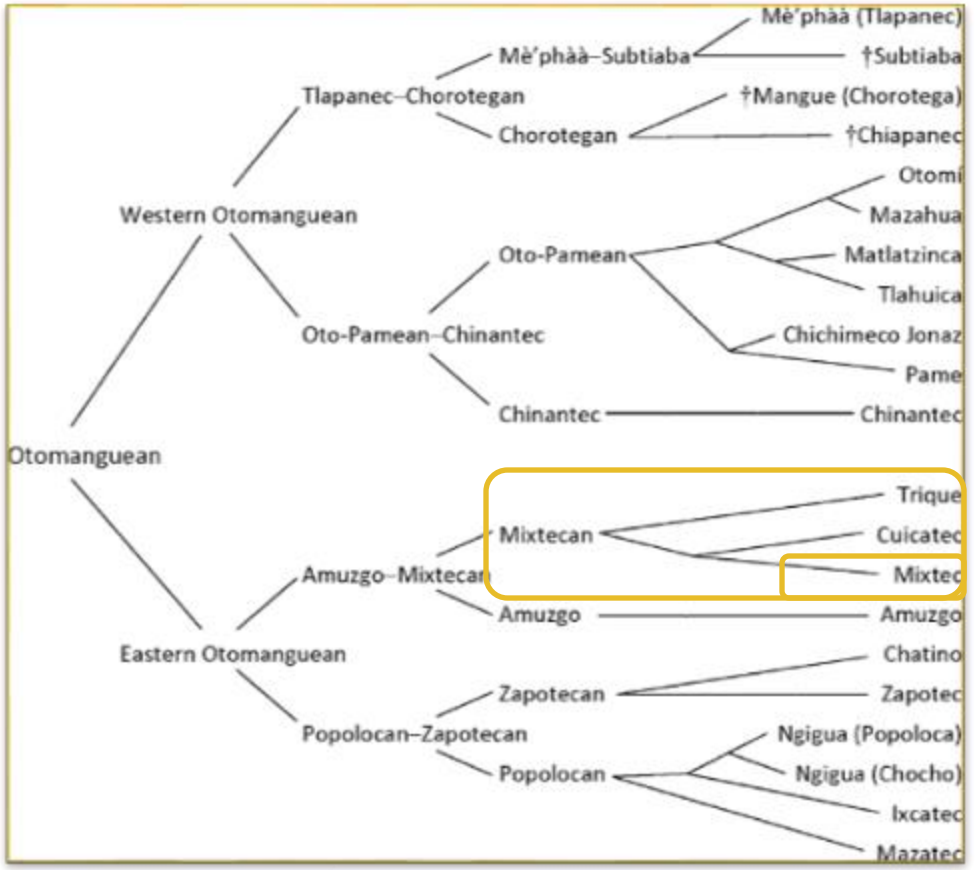
The diversification of Mixtec





The Mixtecan language family



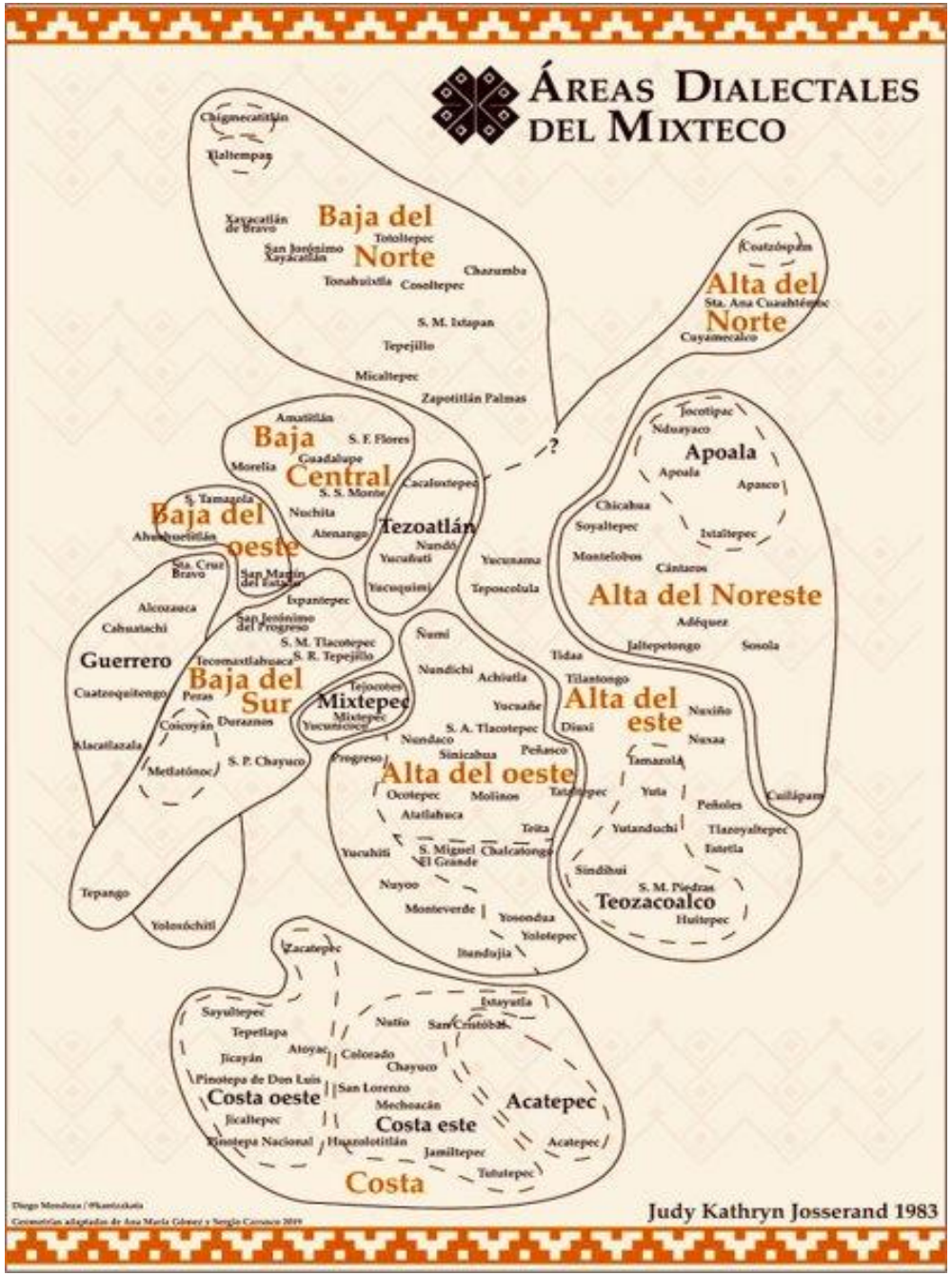


Rieger 2018. Boletín de Antropología. 33 (56) | Campbell 2017 based on Kaufman 1988

Internal diversity of Mixtec

- ~200 local varieties, many of them mutually unintelligible
- traditionally characterized as a dialect continuum
- division into 12 dialect areas with further subdivisions (Josserand 1983)
  - classification based on isoglosses of vowel correspondences

How did we get here?



Josserand 1983, adapted by Diego Mendoza

Subgrouping and classification

"El problema es la aparente violación o infracción de las expectativas ideales del método comparativo en la linguística histórica (...). Pero este modelo presupone una efectiva separación de los grupos, después de un cambio que los divide. En contraste, en la Mixteca (...) vemos una situación más dinámica en el desarrollo de las familias lingüísticas.'' (Bradley & Josserand 1982:303)

English: *The problem is the apparent violation or infringement of the expectations and ideals of the comparative method in historical linguistics (...). But this model assumes an actual separation of the groups, once a change has divided them. In contrast, in the Mixteca, we see a more dynamic situation concerning the development of the language families.*

Bradley, H. C., & Josserand, K. J. 1982. El protomixteco y sus descendientes. *Anales de Antropología* 19(2)

Diversification and the Comparative Method

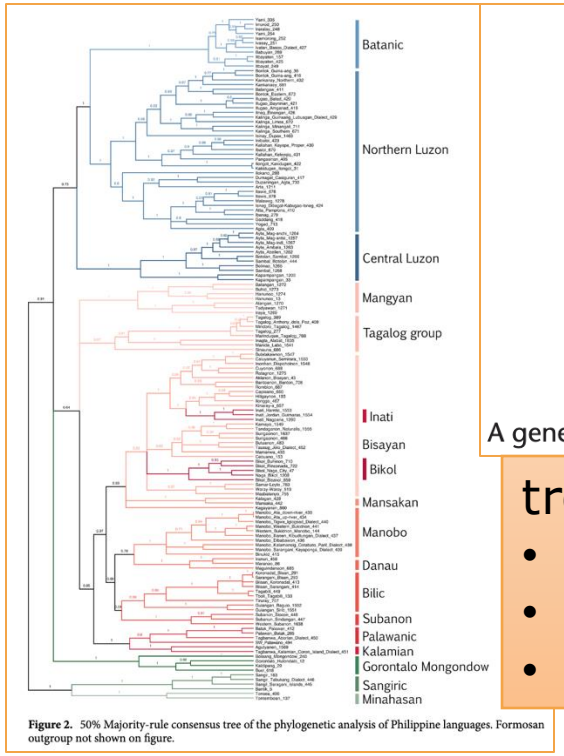
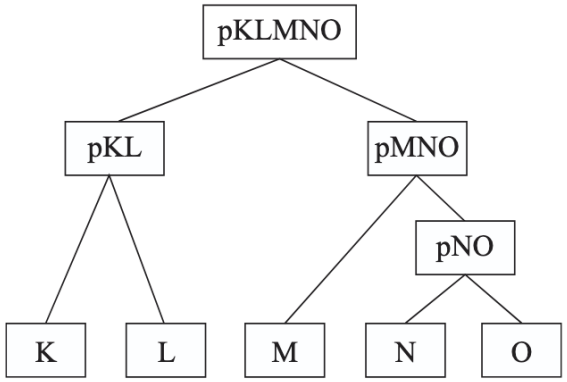


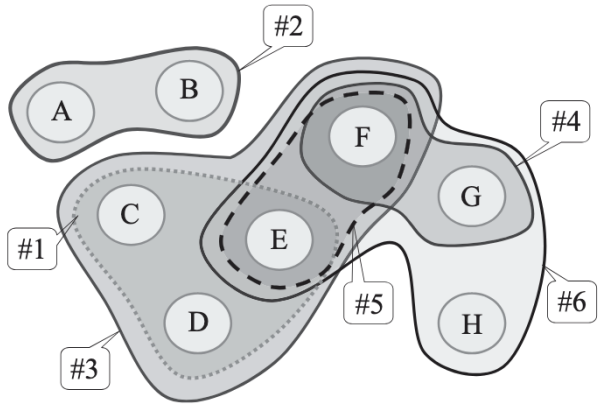
Figure 2. 50% Majority-rule consensus tree of the phylogenetic analysis of Philippine languages. Formosan outgroup not shown on figure.



A genealogical tree indicating internal subgrouping

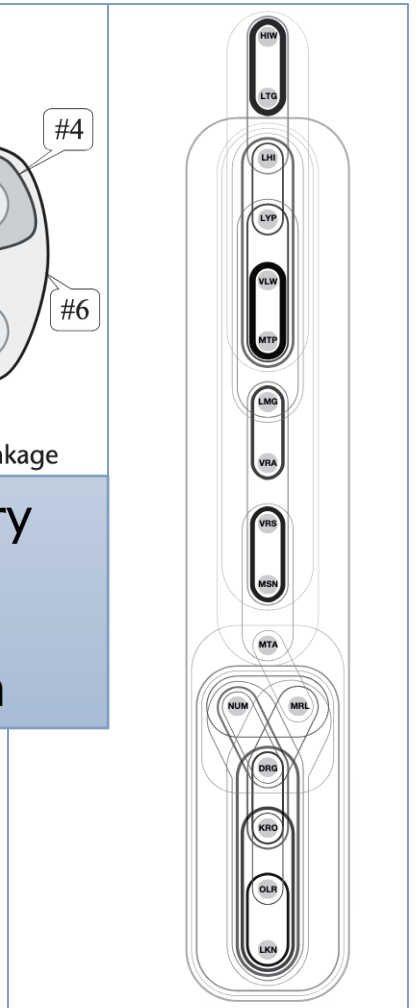
tree model / phylogenetics
- splits and separation
- languages
- time-depth, uncertainty

Comparative Method



Intersecting isoglosses in a dialect continuum or a linkage

wave model / glottometry
- ongoing contact
- dialects & linkages
- strength of interaction

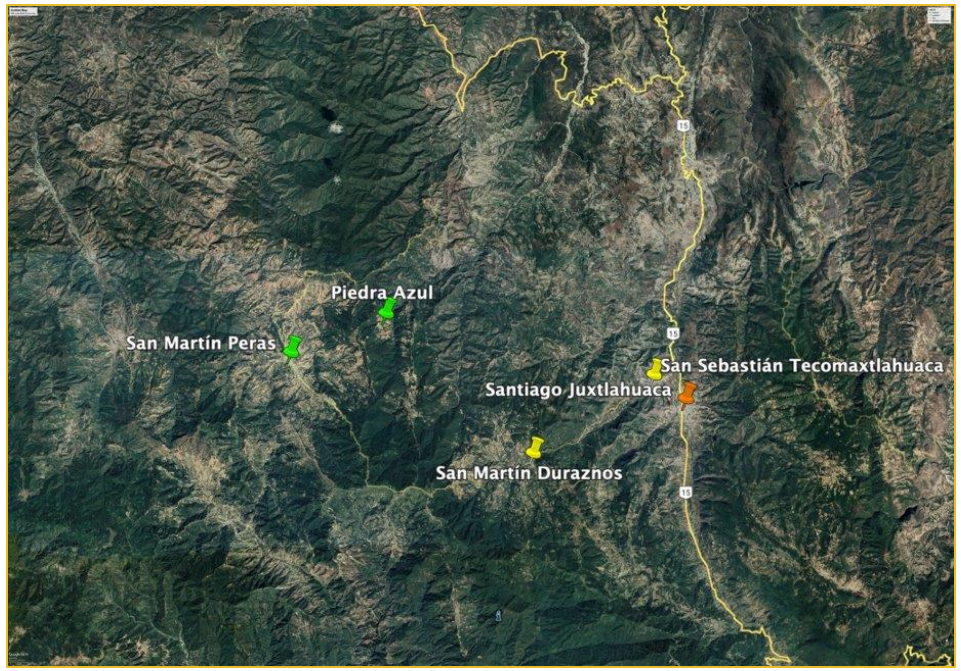


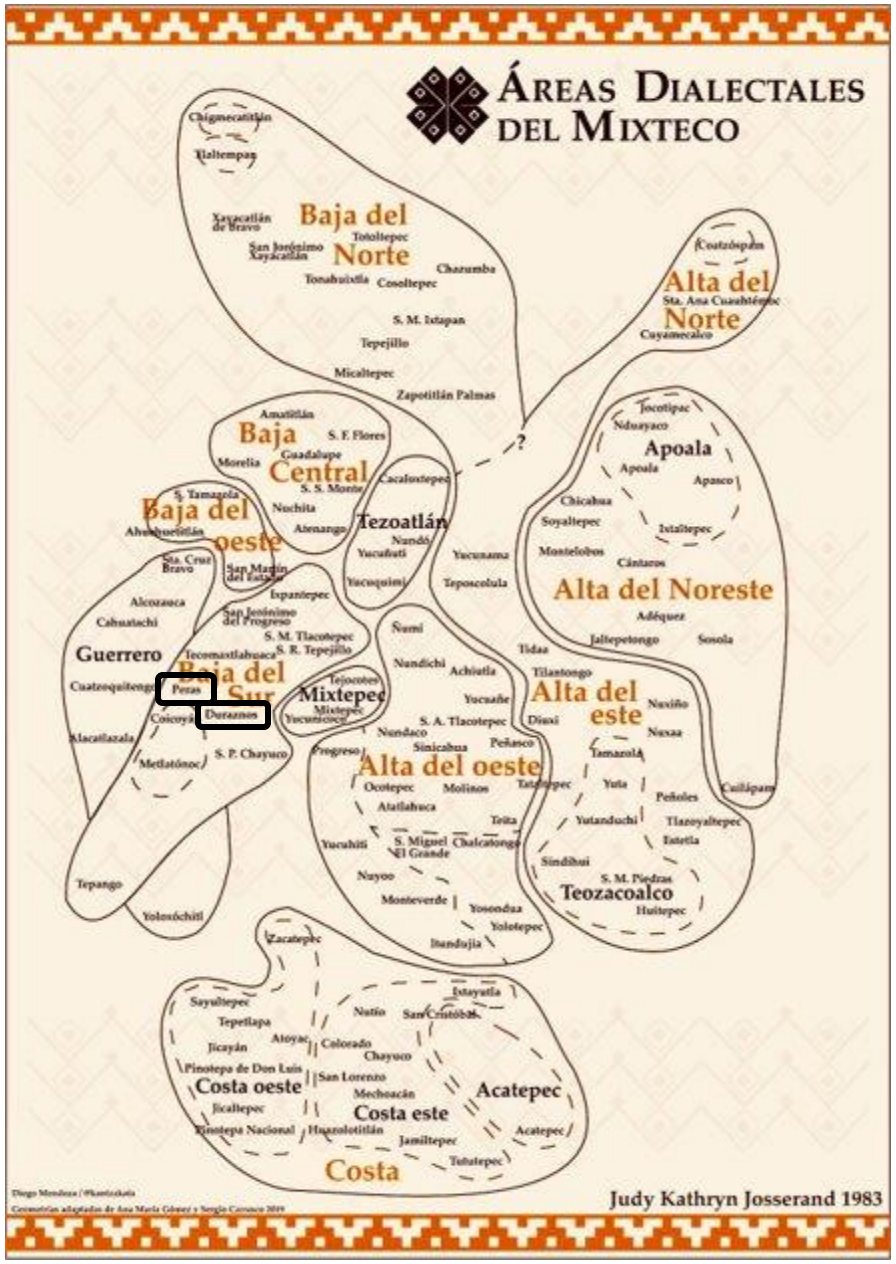
A glottometric diagram of the Torres–Banks languages

François, A. 2015. Trees, waves and linkages: Models of language diversification. *The Routledge Handbook of Historical Linguistics* | King, B. et al. 2024. Bayesian phylogenetic analysis of Philippine languages supports a rapid migration of Malayo-Polynesian languages. *Scientific Reports*, *14*(1)

Case study

- San Martín Duraznos Mixtec
- San Martín Peras Mixtec – Piedra Azul variety





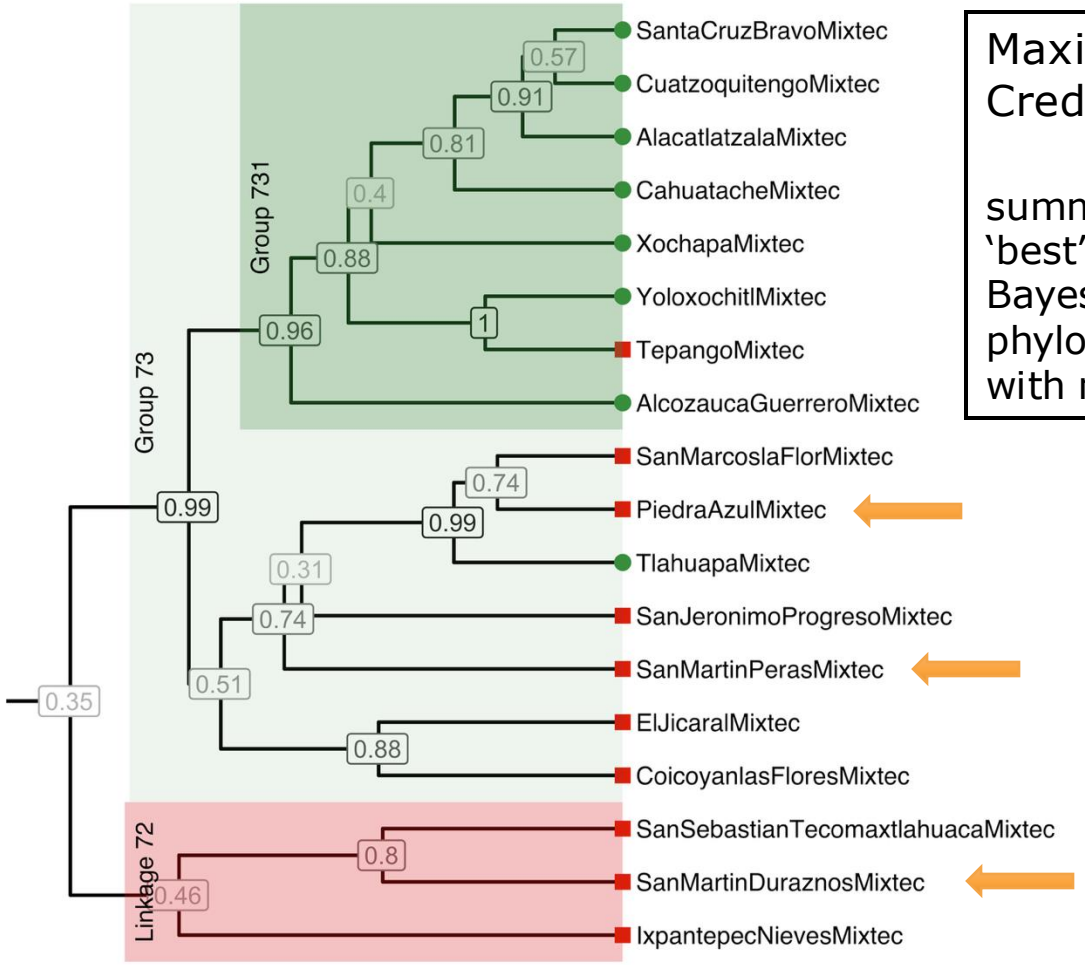
Josserand 1983, adapted by Diego Mendoza

Cognacy and language in use in Peras and Duraznos Mixtec

cognate vocabulary: 83% cognate in 209-item basic vocabulary list

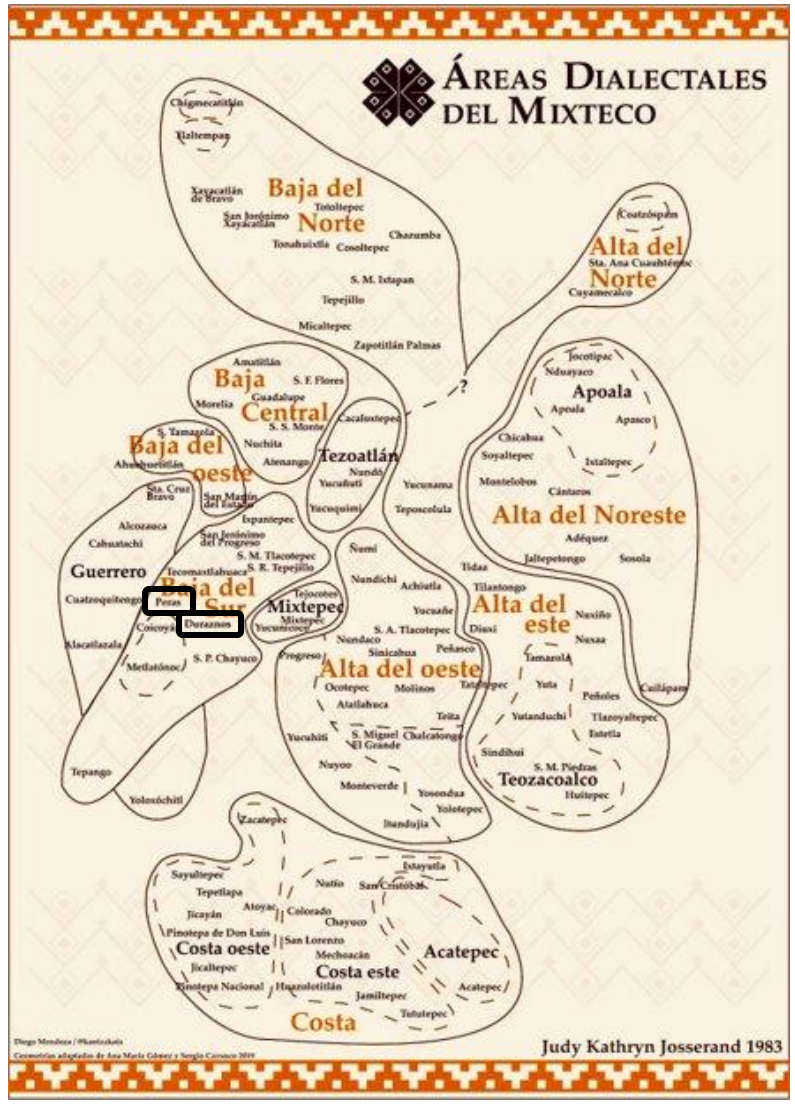
	FISH	BIG.SG	WAX	SISTER.F	STOOL
SMP	tsʲ a ¹ k a ⁵	k a ⁵ ʔ n u ³	i ¹ h m a ¹⁵	k u ³ ʔ u ¹	tsʲ a ⁵ j i ³
SMD	tɕ a ³ k a ⁵	k a ⁵ ʔ n u ³	ɲ u ¹ m a ¹	k i ¹ ʔ β i ¹	tɕ e ³ i ¹

Classification of Duraznos and Peras Mixtec



Maximum Clade Credibility Tree

summary of the 'best' trees of a Bayesian phylogenetic analysis with node posteriors



Auderset, S. et al. 2023. Subgrouping in a 'dialect continuum': A Bayesian phylogenetic analysis of the Mixtecan language family. *Journal of Language Evolution*, *8*(1)

Cognacy and language in use in Peras and Duraznos Mixtec

cognate vocabulary: 83% cognate in 209-item basic vocabulary list

	FISH	BIG.SG	WAX	SISTER.F	STOOL
SMP	tsʲa¹ka⁵	ka⁵ʔnu³	i¹hma¹⁵	ku³ʔ u¹	tsʲa⁵ji³
SMD	tɕa³ka⁵	ka⁵ʔnu³	ɲu¹ ma¹	ki¹ ʔβi¹	tɕe³ i¹

Greeting (to a woman, respectful)

SMP		
i⁵jo³	ũ⁵	ʃi³ʃi³
IPFV.exist	2SG.NHON	aunt

SMD		
tʃa⁵a³	ni⁵	ʃi³ʃi³
greeting	2SG.HON	aunt

Thanks (to a peer, informal)

SMP	
ta³ʃa¹ʔβi³	ũ⁵
thanks	2SG.NHON

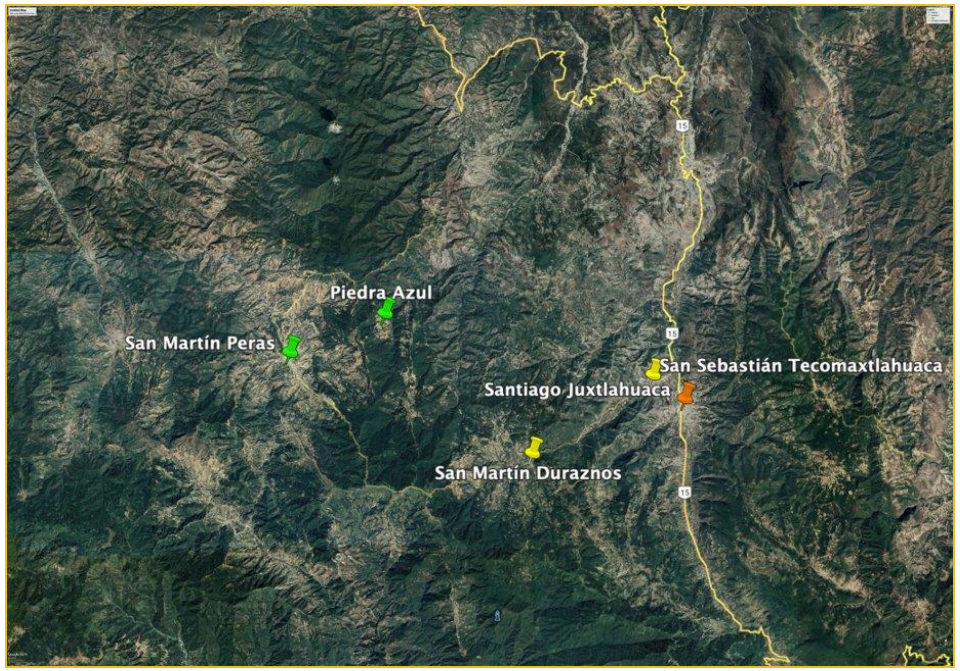
SMD		
ʃi¹nte¹e⁵	i³ni³	ũ³
thanks	inside	2SG.NHON

Duraznos and Peras Mixtec today

San Martín Duraznos (SMD)
~400 speakers in Mexico

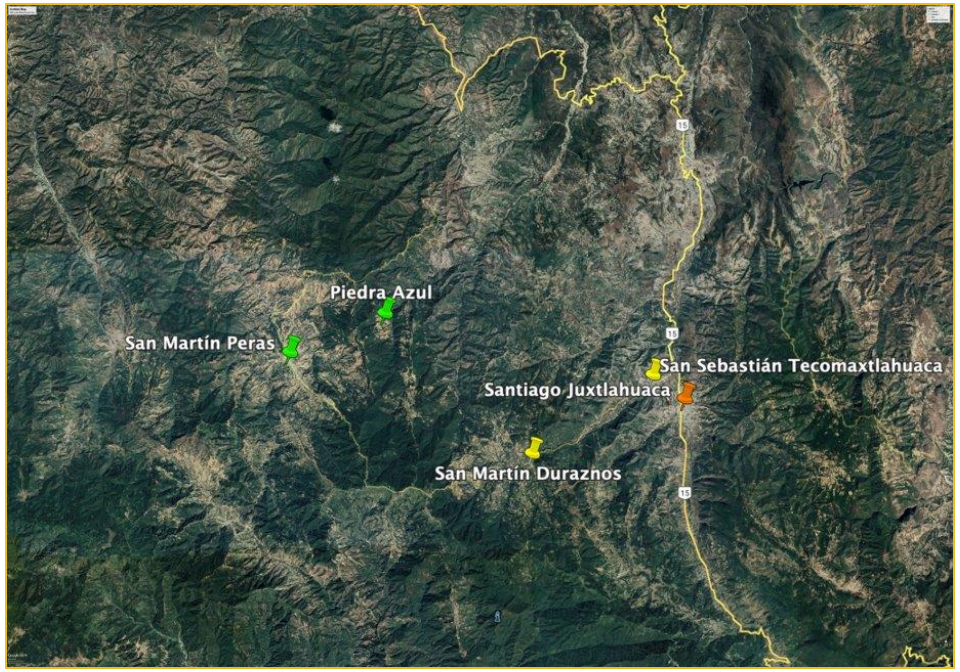
San Martín Peras (SMP)
~ 11,500 speakers in Mexico

- geographic proximity: 25km apart
- orient toward different commercial centers, belong to separate political districts
- relatively little (sustained) contact
- not (readily) mutually intelligible



Duraznos and Peras Mixtec in the past

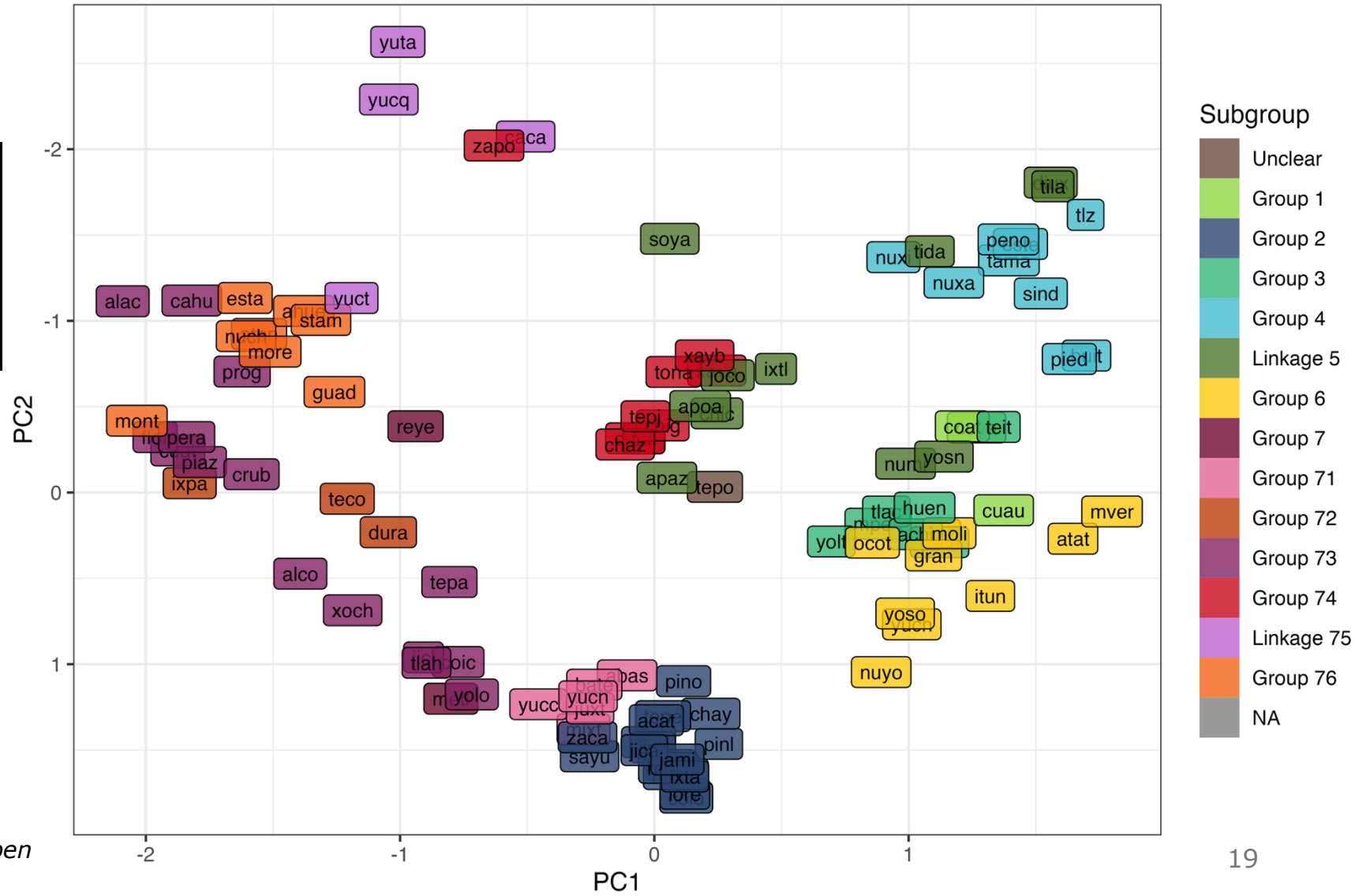
- San Martín Peras, Juxtlahuaca and Tecomaxtlahuaca already existed in the (early) colonial period
- Juxtlahuaca served as the colonial center of area
- San Martín Duraznos was established only around 1860 by people from the Tecomaxtlahuaca area – the town moved at least twice



Mixtec sound changes

Principal Component Analysis of sound changes

the closer together the languages are, the more changes they share



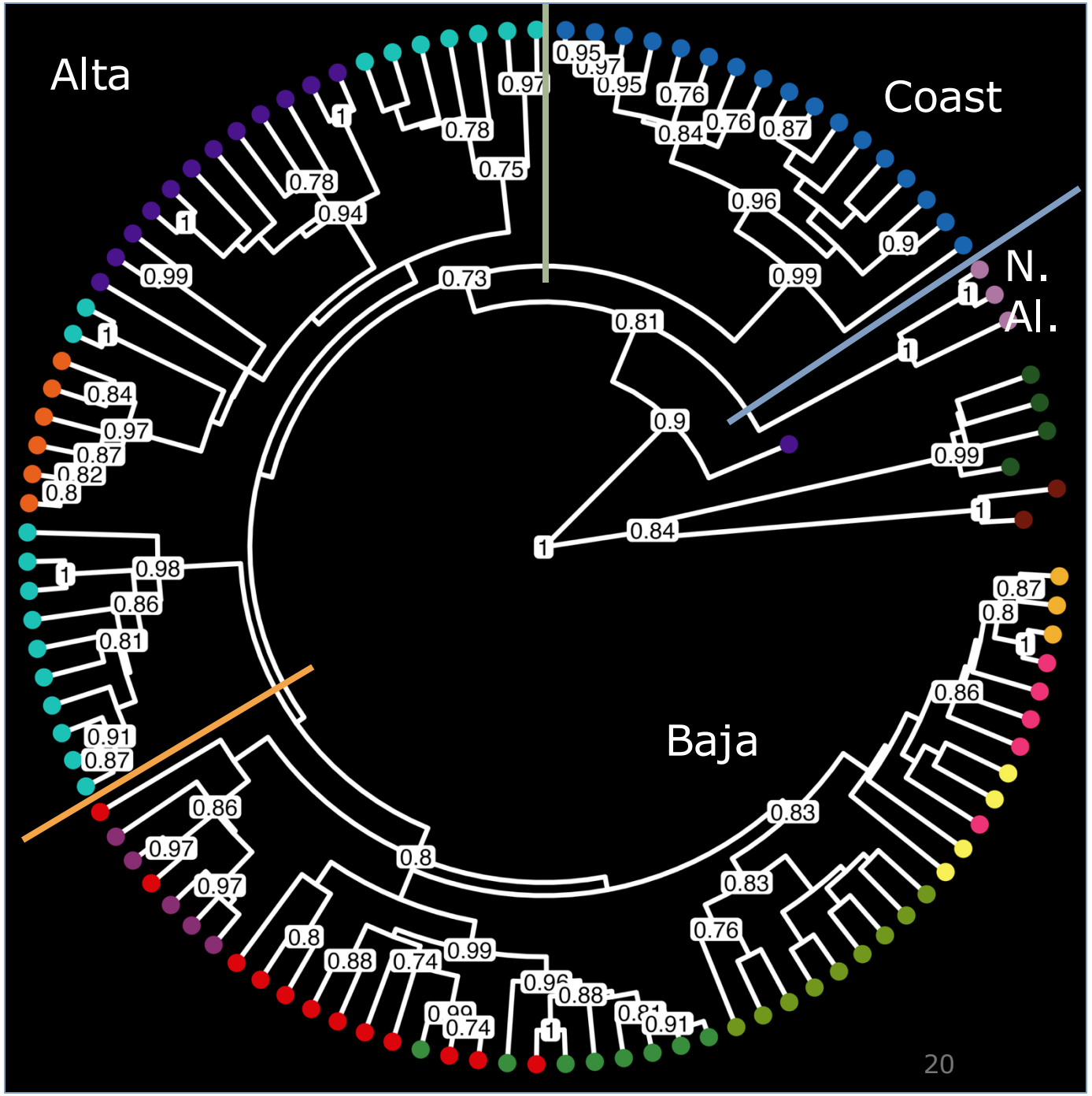
Auderset, S., & Campbell, E. W. (2024). A Mixtec sound change database. *Journal of Open Humanities Data*, *10*(1)

Towards integration

Maximum Clade Credibility Tree

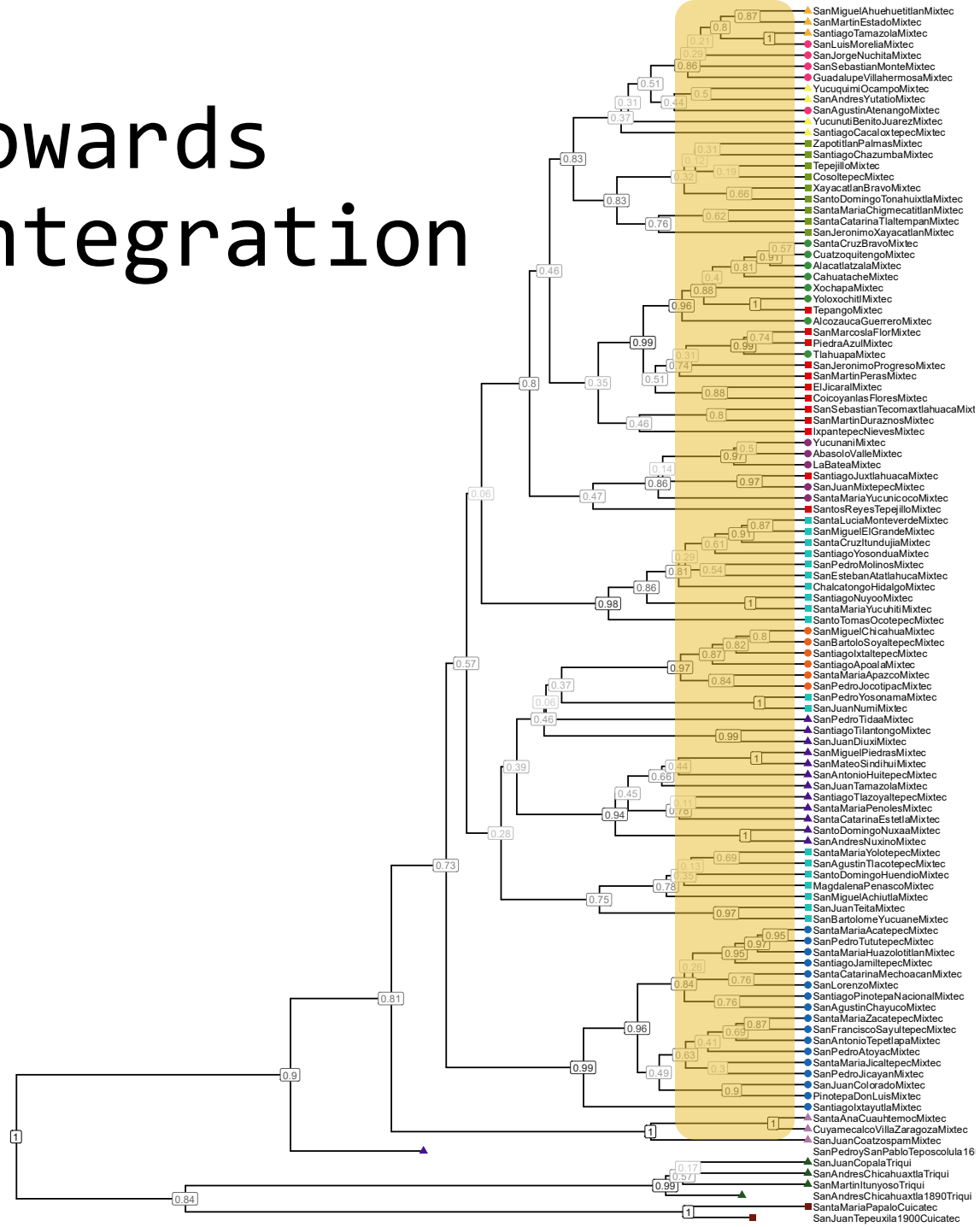
summary of the 'best' trees of a Bayesian phylogenetic analysis with node posteriors

- early migrations to the north and the coast (~1000 CE)
- repeated migrations of whole villages to new locations (throughout (pre-)history)
- during the Colonial period, many people retreated to higher elevations/more remote areas



Auderset, S. et al. 2023. *Journal of Language Evolution*, *8*(1) |
Josserand, J. 1983. *Mixtec dialect history.* U of Tulane PhD Thesis

Towards integration



Maximum Clade Credibility Tree

summary of the ‘best’ trees of a Bayesian phylogenetic analysis with node posteriors

Hypothesis: recent diversification (partially) due to colonization

Auderset, S. et al. 2023. Subgrouping in a ‘dialect continuum’: A Bayesian phylogenetic analysis of the Mixtecan language family. *Journal of Language Evolution*, *8*(1)

Some concluding thoughts

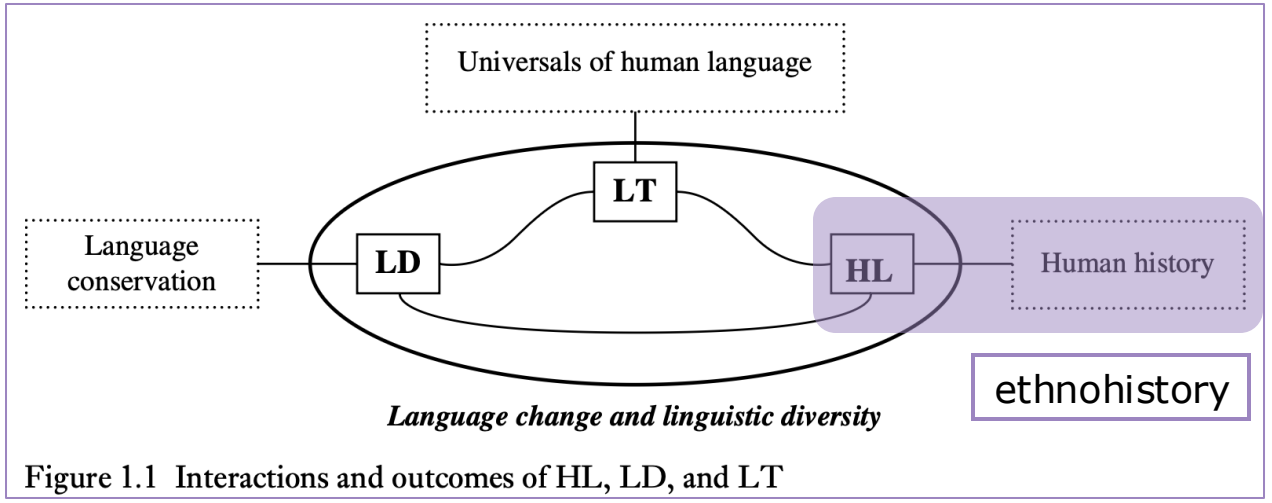


Figure 1.1 Interactions and outcomes of HL, LD, and LT

- colonization can lead to diversification (Ecuadorian Quechua, Mixtec, …)
- diversification cannot be understood through purely linguistic approaches
  - ethnohistory and social dynamics are integral parts of interpretation
- before we explore 'deep history' we should first try to understand 'shallow history'
  - the Americas might be a good starting point because of their shared colonial history

Chacon, T. C. et al. 2022. Language Change and Diversity at the Crossroads of Historical Linguistics, Language Documentation, and Linguistic Typology. *Language change and linguistic diversity*

Shìntèé ini-ntò!

Taxà'vi-ntó!

Thank you!

Acknowledgements

Inî G. Mendoza

Simon L. Peters

Carmen Hernández Martínez

Catalina Martínez Ramírez

Griselda Reyes Basurto

Eric W. Campbell

Mixteco/Indígena Community Organizing Project (MICOP)

ELDP (SG5066)

NSF award #1660355 to UCSB